Driver Drowsiness and Distraction Detection System using Computer Vision

1.Engineering Design Processes – Process sheet

REQUIREMENT ENGINEERING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify Need	Road accident statistics, driver safety issues	Literature Survey, Observation	Driver Safety, Accuracy	Problem Statement
Plan – Gather Requirements	User needs, project objectives	Survey, Journal Papers, Discussion	Detection Accuracy, Real-time Performance	Requirement List
Do – Define Specifications	Requirement List	SRS Preparation	Eye Detection, Yawning Detection, Phone Detection	System Specifications
Check – Validate Requirements	SRS Document	Requirement Review	Completeness, Feasibility	Approved SRS
Act – Refine Requirements	Feedback	Requirement Analysis	Reliability, Performance	Updated SRS
Act – Freeze Requirements	Final SRS	Approval	Stability	Requirement Freeze

CONCEPT DESIGN

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Generate Concepts	Requirements	Brainstorming	Driver Safety, AI Integration	Concept Ideas
Plan – Compare Concepts	Concept Ideas	Pugh Matrix	Accuracy, Processing Speed	Selected Concept
Do – Develop System Architecture	Selected Concept	Functional Block Diagram	Module Integration	System Architecture
Check – Evaluate Concept	Architecture	Design Review	Feasibility, Reliability	Evaluated Concept
Act – Improve Concept	Review Feedback	Refinement	Accuracy, Response Time	Improved Concept
Act – Freeze Concept	Final Concept	Approval	Reliability	Concept Freeze

EMBODIMENT DESIGN

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Define Architecture	Concept	System Design	AI Module Integration	System Architecture
Plan – Select Components	Laptop, Webcam	Datasheets, Comparison	Compatibility	Component List
Do – Design Software Architecture	Components	OpenCV, MediaPipe, YOLOv8 Design	Processing Efficiency	Software Architecture

Check – Verify Design	Software Design	Design Review	Accuracy	Verified Design
Act – Optimize Design	Review Feedback	Optimization	Speed, Reliability	Optimized Design
Act – Freeze Embodiment	Final Design	Approval	Performance	Design Freeze

DETAIL DESIGN

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Define Detail Scope	Architecture	Planning	Functional Modules	Detail Plan
Plan – Specify Components	Hardware & Software	Requirement Analysis	Performance	Specification Sheet
Do – Prepare Detailed Design	Specifications	Python, OpenCV, MediaPipe, YOLOv8	Detection Accuracy	Detailed Design
Check – Review Design	Detailed Design	Design Review	Completeness	Review Report
Act – Correct Design	Review Feedback	Modification	Compliance	Updated Design
Act – Release Design	Final Design	Approval	Functionality	Design Release

SOURCING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Prepare Software & Hardware List	Final Design	BOM Preparation	Completeness	Component List
Plan – Select Components	BOM	Vendor Comparison	Cost, Availability	Approved Components
Do – Install Software Packages	Approved List	Python, pip	Version Compatibility	Installed Environment
Check – Verify Installation	Installed Software	Functional Testing	Compatibility	Installation Report
Act – Resolve Issues	Installation Report	Reinstallation	Stability	Working Environment
Act – Approve Setup	Final Setup	Validation	Reliability	Setup Complete

PROTOTYPE BUILDING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Build Plan	**Approved Design**	**Planning**	**Development Sequence**	**Build Plan**
Plan – Prepare Setup	Camera, Laptop	Checklist	Availability	Ready Setup
Do – Develop Detection System	Software Modules	Python Programming	EAR, MAR, YOLO Accuracy	Prototype V1
Check – Verify Prototype	Prototype	Functional Testing	Detection Accuracy	Test Report

Act – Fix Errors	Test Report	Debugging	Reliability	Prototype V2
Act – Finalize Prototype	Updated Prototype	Validation	Performance	Final Prototype

PROTOTYPE TESTING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Prepare Test Plan	Final Prototype	Test Planning	Coverage	Test Plan
Plan – Setup Testing Environment	Prototype	Camera Calibration	Lighting	Test Setup
Do – Perform Functional Testing	Prototype	Live Driver Testing	EAR, MAR, YOLO Detection	Test Data
Check – Analyze Results	Test Data	Performance Analysis	Accuracy, Precision	Test Report
Act – Identify Issues	Test Report	Root Cause Analysis	False Detection	Improvement List
Act – Approve Testing	Final Results	Validation	Acceptance	Test Approval

ITERATION & OPTIMIZATION

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify Issues	Test Report	Analysis	Detection Errors	Improvement Plan
Plan – Select Improvements	Improvement Areas	Decision Matrix	Accuracy	Optimization Strategy

Do – Implement Changes	Existing System	Parameter Tuning	Detection Speed	Improved System
Check – Retest System	Updated System	Functional Testing	Reliability	Test Report
Act – Fine Tune System	Test Results	Hyperparameter Tuning	Stability	Optimized System
Act – Freeze Optimized Design	Final System	Approval	Performance	Final Design

DOCUMENTATION & SHARING

Process Step	Input	Tools / Methods	Critical Parameters	Output
Plan – Identify Documents	Project Data	Planning	Completeness	Document List
Plan – Organize Data	Design Files, Test Results	Documentation	Clarity	Organized Data
Do – Prepare Report	Project Information	MS Word, Flowcharts	Accuracy	Draft Report
Check – Review Documentation	Draft Report	Proofreading	Consistency	Review Report
Act – Correct Documentation	Review Feedback	Editing	Quality	Final Report
Act – Submit Project	Final Documents	Presentation & Submission	Compliance	Project Completion

2.ACTIVITY DURATION TABLE (250 HOURS)

S.No	Activity Description	Duration (Hours)
1	Identify driver safety problems	4
2	Collect user requirements	5
3	Study AI, Computer Vision, MediaPipe and YOLOv8	5
4	Define system requirements	5
5	Review requirements	4
6	Freeze requirements	3
7	Generate concept ideas	5
8	Compare concepts	5
9	Develop system block diagram	5
10	Evaluate concepts	5
11	Improve concept	4
12	Freeze concept design	3
13	Define system architecture	5
14	Select hardware components	5
15	Select software tools	5
16	Design software workflow	5
17	Verify component compatibility	5
18	Freeze embodiment design	4

19	Define detailed design scope	5
20	Prepare software specifications	5
21	Develop face detection module	7
22	Develop eye and mouth detection module	7
23	Develop head position detection module	6
24	Develop mobile phone detection module (YOLOv8)	7
25	Prepare hardware and software setup	4
26	Install required libraries and packages	4
27	Integrate webcam with software	5
28	Implement alert system	5
29	Implement CSV event logging	4
30	Implement screenshot capture	4
31	Integrate all software modules	6
32	Prepare test plan	4
33	Setup testing environment	4
34	Test drowsiness detection	5
35	Test yawning detection	5
36	Test head position detection	5
37	Test mobile phone detection	5
38	Test real-time monitoring	5

39	Verify CSV logging and screenshots	4
40	Analyze test results	4
41	Identify system improvements	4
42	Optimize detection accuracy	5
43	Improve YOLOv8 detection performance	5
44	Improve MediaPipe tracking accuracy	5
45	Retest optimized system	5
46	Freeze optimized design	3
47	Prepare project documentation	5
48	Organize screenshots and results	4
49	Prepare final report	5
50	Review documentation	3
51	Final corrections	4
52	Prepare presentation (PPT)	4
53	Final system demonstration	3
54	Project submission	2
TOTAL		250

3. ACTIVITY PRECEDENCE TABLE

Activity No.	Activity Description	Immediate Predecessor
1	Identify driver safety problems	—
2	Collect user requirements	1
3	Study AI, Computer Vision, MediaPipe and YOLOv8	2
4	Define system requirements	3
5	Review requirements	4
6	Freeze requirements	5
7	Generate concept ideas	6
8	Compare concepts	7
9	Develop system block diagram	8
10	Evaluate concepts	9
11	Improve concept	10
12	Freeze concept design	11
13	Define system architecture	12
14	Select hardware components	13
15	Select software tools	14
16	Design software workflow	15
17	Verify component compatibility	16

18	Freeze embodiment design	17
19	Define detailed design scope	18
20	Prepare software specifications	19
21	Develop face detection module	20
22	Develop eye and mouth detection module	21
23	Develop head pose detection module	22
24	Develop mobile phone detection module (YOLOv8)	23
25	Prepare hardware and software setup	24
26	Install required libraries and packages	25
27	Integrate webcam with software	26
28	Implement alert system	27
29	Implement CSV event logging	28
30	Implement screenshot capture	29
31	Integrate all software modules	30
32	Prepare test plan	31
33	Setup testing environment	32
34	Test drowsiness detection	33
35	Test yawning detection	34
36	Test head position detection	35
37	Test mobile phone detection	36

38	Test real-time monitoring	37
39	Verify CSV logging and screenshots	38
40	Analyze test results	39
41	Identify system improvements	40
42	Optimize detection accuracy	41
43	Improve YOLOv8 detection performance	42
44	Improve MediaPipe tracking accuracy	43
45	Retest optimized system	44
46	Freeze optimized design	45
47	Prepare project documentation	46
48	Organize screenshots and results	47
49	Prepare final report	48
50	Review documentation	49
51	Final corrections	50
52	Prepare presentation (PPT)	51
53	Final system demonstration	52
54	Project submission	53

Activity Listing

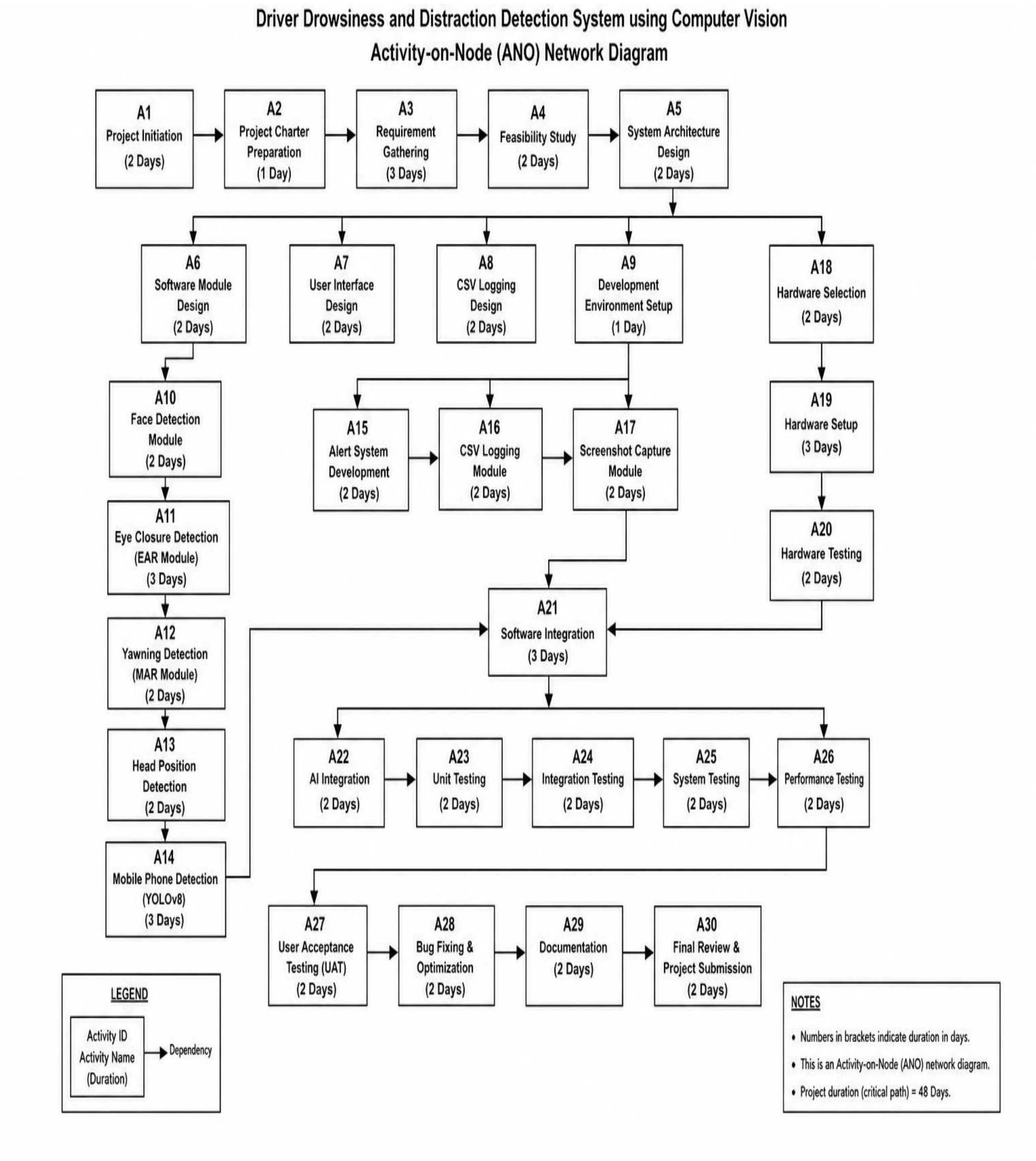
Activity ID	Activity Name	Description	Duration (Days)	Predecessor (s)	Responsible Role
A1	Project Initiation	Define project objectives, scope and team responsibilities	2	-	Project Manager
A2	Project Charter Preparation	Prepare and approve project charter	1	A1	Project Manager
A3	Requirement Gathering	Collect functional and non-functional requirements	3	A2	Business Analyst
A4	Feasibility Study	Analyze technical and operational feasibility	2	A3	Project Manager
A5	System Architecture Design	Design overall AI-based system architecture	2	A4	System Designer
A6	Software Module Design	Design OpenCV, MediaPipe and YOLOv8 modules	2	A5	AI Developer

A7	User Interface Design	Design monitoring dashboard and alert interface	2	A5	UI/UX Designer
A8	Database & CSV Logging Design	Design CSV logging and event recording system	2	A5	Software Developer
A9	Development Environment Setup	Install Python, OpenCV, MediaPipe and YOLOv8	1	A5	Software Developer
A10	Face Detection Module	Develop driver face detection module	2	A9	Software Developer
A11	Eye Closure Detection Module	Develop EAR-based eye monitoring module	3	A10	AI Developer
A12	Yawning Detection Module	Develop MAR-based yawning detection	2	A11	AI Developer
A13	Head Position Detection Module	Develop head pose estimation module	2	A12	AI Developer
A14	Mobile Phone Detection Module	Develop YOLOv8 mobile phone detection	3	A13	AI Developer

A15	Alert System Development	Develop warning message and alarm module	2	A14	Software Developer
A16	CSV Logging Module	Develop CSV event logging with date and time	2	A15	Software Developer
A17	Screenshot Capture Module	Implement automatic screenshot capture	2	A16	Software Developer
A18	Hardware Selection	Select laptop/webcam and required peripherals	2	A5	Hardware Engineer
A19	Hardware Setup	Configure webcam and hardware components	3	A18	Hardware Engineer
A20	Hardware Testing	Verify webcam and hardware functionality	2	A19	Hardware Engineer
A21	Software Integration	Integrate all software modules	3	A17, A20	Software Developer
A22	AI Model Integration	Integrate OpenCV, MediaPipe and YOLOv8	2	A21	AI Developer
A23	Unit Testing	Test individual modules	2	A22	Testing Engineer

A24	Integration Testing	Verify interaction between all modules	2	A23	Testing Engineer
A25	System Testing	Perform complete driver monitoring system testing	2	A24	Testing Engineer
A26	Performance Testing	Evaluate detection speed, accuracy and reliability	2	A25	Testing Engineer
A27	User Acceptance Testing (UAT)	Validate system using real driving scenarios	2	A26	End User & Project Manager
A28	Bug Fixing & Optimization	Resolve defects and optimize performance	2	A27	Development Team
A29	Documentation	Prepare technical documentation and user manual	2	A28	Documentation Engineer
A30	Final Review & Project Submission	Final verification, presentation and submission	2	A29	Project Manager

Network Diagram



Critical Path Method (CPM) Calculation

Act. ID	Activity Name	Dur. (Days)	Predecessor(s)	ES	EF	LS	LF	Total Float	Critical
A1	Project Initiation	2	-	0	2	0	2	0	Yes
A2	Project Charter Preparation	1	A1	2	3	2	3	0	Yes
A3	Requirement Gathering	3	A2	3	6	3	6	0	Yes
A4	Feasibility Study	2	A3	6	8	6	8	0	Yes
A5	System Architecture Design	2	A4	8	10	8	10	0	Yes
A6	Software Module Design	2	A5	10	12	11	13	1	No
A7	User Interface Design	2	A5	10	12	18	20	8	No
A8	CSV Logging Design	2	A5	10	12	18	20	8	No
A9	Development Environment Setup	1	A5	10	11	10	11	0	Yes
A10	Face Detection Module	2	A9	11	13	11	13	0	Yes
A11	Eye Closure Detection Module	3	A10	13	16	13	16	0	Yes

A12	Yawning Detection Module	2	A11	16	18	16	18	0	Yes
A13	Head Position Detection	2	A12	18	20	18	20	0	Yes
A14	Mobile Phone Detection	3	A13	20	23	20	23	0	Yes
A15	Alert System Development	2	A14	23	25	23	25	0	Yes
A16	CSV Logging Module	2	A15	25	27	25	27	0	Yes
A17	Screenshot Capture Module	2	A16	27	29	27	29	0	Yes
A18	Hardware Selection	2	A5	10	12	20	22	10	No
A19	Hardware Setup	3	A18	12	15	22	25	10	No
A20	Hardware Testing	2	A19	15	17	25	27	10	No
A21	Software Integration	3	A17, A20	29	32	29	32	0	Yes
A22	AI Integration	2	A21	32	34	32	34	0	Yes
A23	Unit Testing	2	A22	34	36	34	36	0	Yes
A24	Integration Testing	2	A23	36	38	36	38	0	Yes
A25	System Testing	2	A24	38	40	38	40	0	Yes
A26	Performance Testing	2	A25	40	42	40	42	0	Yes

A27	User Acceptance Testing	2	A26	42	44	42	44	0	Yes
A28	Bug Fixing & Optimization	2	A27	44	46	44	46	0	Yes
A29	Documentation	2	A28	46	48	46	48	0	Yes
A30	Final Review & Project Submission	2	A29	48	50	48	50	0	Yes

Critical Path

The activities with zero total float form the Critical Path:

A1 → A2 → A3 → A4 → A5 → A9 → A10 → A11 → A12 → A13 → A14 → A15 → A16 → A17 → A21 → A22 → A23 → A24 → A25 → A26 → A27 → A28 → A29 → A30

These activities must be completed on schedule because any delay in these activities will directly affect the overall project completion time.

Project Summary

Parameter	**Value**
Total Activities	30
Project Duration (Planned)	50 Days
Critical Activities	24
Non-Critical Activities	6
Maximum Float	10 Days (Hardware & UI Design Activities)

Minimum Float	0 Days
Scheduling Technique	Critical Path Method (CPM)